RetailCore

Retail Sales & Operations Data Management System

Milestone 1: Dataset Identification & Tentative Analytical Queries

Course	DS604 - Introduction to Data Management (Autumn 2026)
Dataset	Retail Data Warehouse - 12 Table 1M+ Rows Dataset (Kaggle)
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This document presents the proposed dataset and tentative analytical queries for Milestone 1 of the DS604 Introduction to Data Management group project. The selected dataset is the **Retail Data Warehouse - 12 Table 1M+ Rows Dataset** published on Kaggle. Public documentation and public projects built on this dataset identify twelve source tables: categories, customers, employees, order_items, orders, payments, products, promotions, returns, shipments, stores and suppliers. The dataset represents a retail business with customers, stores, employees, suppliers, a product catalogue, sales orders, payments, shipments, promotions and product returns.

> **Course requirement alignment:** the supplied project brief asks students to identify a dataset with at least around five tables and submit a document describing the dataset and listing analytical queries. Subsequent milestones cover the ER diagram, relational schema, DDL, PostgreSQL loading using Pandas/SQLAlchemy, analytical SQL, and views.

1. Dataset Overview

Attribute	Details
Domain	Retail sales, product catalogue, supply chain and store operations.
Nature	Training-oriented, warehouse-style data published on Kaggle by Datarspectrum Technology Training Center. The data uses numeric identifiers and contains no personal names or contact details, which indicates that it is synthetic/simulated rather than data from a real retailer.
Scale	Twelve CSV files (about 14 MB in total on the Kaggle listing) and more than one million rows overall. Public analyses report roughly 1.5 million rows in total, with order_items (about 600,000 rows) as the largest table, orders at about 300,000 rows and customers at about 50,000 rows. The exact row count of every table will be recorded from the team's downloaded copy before final submission.
Time context	Public profiling reports order dates spanning 01 January 2020 to 01 January 2024 (about four years).

Why this dataset fits the project

- **Multi-table structure:** documented as twelve related tables, well above the course requirement of around five.
- **Relational design potential:** a clean star/snowflake-like structure with a central fact table (order_items), eleven foreign-key relationships, 1:1 relationships (payments and shipments to orders) and optional 0..* relationships (orders without line items, customers without orders).
- **Analytical SQL potential:** supports joins across many tables, grouping, HAVING, subqueries/CTEs, aggregates, ranking and window functions, date-time analysis, anti-joins and views.
- **Manageable scale:** about 14 MB of CSV and a largest table of about 600,000 rows can be loaded into PostgreSQL with Pandas to_sql in append mode (chunked inserts) without special infrastructure.
- **Data-management relevance:** multiple parent/child tables make key design, referential integrity, load ordering, normalization and query optimization meaningful, and the data allows realistic data-quality checks.

2. Proposed Source Tables

The columns below are those observed in public implementations of this dataset. They will be re-confirmed against the downloaded CSV files before Milestone 2 so that no column, key or relationship is invented without evidence.

Source table	Purpose	Columns (to be verified)
categories.csv	Product classification	category_id (PK), category_name. About 30 distinct categories are used by products.
suppliers.csv	Supplier master	supplier_id (PK), country.
stores.csv	Store master	store_id (PK), city. Public profiling reports only a few distinct store cities.
customers.csv	Customer profile	customer_id (PK), city, signup_date. No names or contact details are present.
promotions.csv	Discount campaigns	promotion_id (PK), discount.
employees.csv	Store staff	employee_id (PK), store_id (FK), salary.
products.csv	Product catalogue	product_id (PK), category_id (FK), supplier_id (FK), price.
orders.csv	Sales orders (header)	order_id (PK), customer_id (FK), store_id (FK), order_date, promotion_id (FK).
order_items.csv	Sales order lines (fact table)	order_item_id (PK), order_id (FK), product_id (FK), qty, price.
payments.csv	Order payments	payment_id (PK), order_id (FK), amount.
shipments.csv	Order shipments	shipment_id (PK), order_id (FK), status.
returns.csv	Product returns/refunds	return_id (PK), order_item_id (FK), refund.

Expected relationships for the later ER design

Child column (FK)	Parent table	Relationship idea
products.category_id	categories	Many products belong to one category.
products.supplier_id	suppliers	Many products are supplied by one supplier.
employees.store_id	stores	Many employees work at one store.
orders.customer_id	customers	A customer places zero or more orders.
orders.store_id	stores	Many orders are placed at one store.
orders.promotion_id	promotions	An order optionally uses one promotion.
order_items.order_id	orders	An order has zero or more line items.
order_items.product_id	products	A product appears in many order lines.
payments.order_id	orders	Payments are linked to an order (public profiling reports one payment per order).
shipments.order_id	orders	Shipments are linked to an order (public profiling reports one shipment per order).
returns.order_item_id	order_items	A return refers to an order line, not directly to a product.

Public profiling reports that all eleven foreign keys resolve with no orphan rows, that about 40,000 of the 300,000 orders have no line items, and that a small number of customers have never placed an order. These observations will be re-verified on the team's copy; they are the reason the order-to-items and customer-to-orders relationships are treated as optional (zero-or-more) in the draft design.

3. Tentative Analytical Queries

The following queries are intentionally written as analytical questions rather than final SQL statements. They are suitable for the PostgreSQL milestone and can be refined after the final relational schema is approved. The list deliberately covers multiple tables and increasingly advanced SQL techniques. **Revenue is defined throughout as the sum of qty × price over order_items**; payments.amount is analysed separately.

ID	Analytical query	Purpose / expected analysis	Main SQL concepts
Q01	Customers by city	Count customers in each city and show the percentage of total customers.	GROUP BY, COUNT, window/percentage
Q02	Customer sign-up trend	Count new customers per year and month using signup_date to see how the customer base grew.	Date truncation, GROUP BY
Q03	Sales by store	For each store (and store city) calculate the number of orders and total revenue, ordered by revenue.	JOIN, SUM, GROUP BY, ORDER BY
Q04	Top products by revenue	Find the top 10 products by total revenue and units sold.	JOIN, SUM, GROUP BY, LIMIT

ID	Analytical query	Purpose / expected analysis	Main SQL concepts
Q05	Category performance	Compare units sold, revenue and number of orders across product categories.	Multiple JOINs, GROUP BY, multiple aggregates
Q06	Supplier revenue by country	Find which supplier countries contribute the most revenue through their products.	Multiple JOINs, SUM, GROUP BY
Q07	Average order value by store city	Calculate total revenue per order first, then the average order value for each store city.	CTE, AVG, JOIN
Q08	Monthly revenue trend	Calculate monthly orders, units sold and revenue across the data period (2020-2024).	DATE_TRUNC, GROUP BY
Q09	Year-over-year revenue growth	Calculate yearly revenue and the percentage change from the previous year.	CTE, LAG window function
Q10	Promotion vs non-promotion orders	Compare order count, revenue and average order value for orders with and without a promotion.	LEFT JOIN, CASE, GROUP BY
Q11	Revenue by discount level	Group orders by promotion discount value and compare order volume and revenue.	JOIN, GROUP BY, AVG, SUM
Q12	Return rate by category	Calculate the share of order lines that were returned for each product category.	JOIN, COUNT, conditional aggregation
Q13	Refunds by supplier country	Find total refund value and number of returns linked to each supplier country, via returns > order_items > products > suppliers.	Multi-table JOIN, SUM, COUNT
Q14	Shipment status distribution	Count shipments by status and show each status as a percentage of all shipments.	GROUP BY, COUNT, percentage
Q15	Orders with missing shipment or payment	Find orders that have no shipment record, or no payment record, to check completeness of the order lifecycle.	LEFT JOIN, IS NULL (anti-join)
Q16	Payment vs order total	Compare payments.amount with the line-item total of each order and list the largest differences.	CTE, JOIN, ABS, ORDER BY
Q17	Repeat customers	Identify customers with more than one order and calculate the repeat-customer rate.	COUNT, GROUP BY, HAVING
Q18	Customer lifetime value	Find the top 10 customers by total spend and show their order counts.	JOIN, SUM, GROUP BY, LIMIT
Q19	Customers who never ordered	List customers who have no orders and summarise them by city and sign-up year.	LEFT JOIN, IS NULL, GROUP BY
Q20	Top customers within each city	Rank customers by total spend within each city and keep the top 3 per city.	Window function, PARTITION BY, RANK
Q21	Top products within each category	Rank products within each category by revenue and keep the top 3 per category.	DENSE_RANK, PARTITION BY
Q22	Employees above store average salary	List employees earning more than the average salary of their own store.	Subquery/CTE, AVG, JOIN

ID	Analytical query	Purpose / expected analysis	Main SQL concepts
Q23	Revenue per employee by store	Compare each store's revenue to its employee count and salary cost (aggregate each table separately to avoid inflated joins).	CTE, multiple aggregates, JOIN
Q24	Time from sign-up to first order	For each customer calculate the days between signup_date and the first order, then summarise the average by city.	MIN, date arithmetic, GROUP BY

4. Proposed Views for Later Milestones

View	Contents
V01 - customer_lifetime_summary	One row per customer: city, sign-up date, number of orders, total units, total spend, first/last order date.
V02 - product_performance_summary	One row per product: category, supplier country, units sold, revenue, number of returns, total refund value.
V03 - monthly_sales_summary	Monthly orders, unique customers, units sold and revenue, with and without promotions.
V04 - store_performance_summary	One row per store: city, number of orders, revenue, average order value, employee count, shipment-status counts.

5. Data Validation Checklist Before Finalizing

1. Record the exact row count of every CSV file and the total number of records.
2. Record the exact column names and data types (in particular, confirm whether price, salary, amount, refund and discount are integers or decimals).
3. Confirm candidate primary keys and whether they are actually unique in every table.
4. Confirm foreign-key coverage for all eleven relationships (e.g. order_items.order_id exists in orders, returns.order_item_id exists in order_items).
5. Measure missing values and duplicate rows in every table, especially optional keys such as orders.promotion_id.
6. Confirm the cardinality of payments and shipments per order (one-to-one or one-to-many) before fixing the ER design.
7. Check orders that have no line items and customers that have no orders, and decide how these optional relationships are modelled.
8. Check date formats and ranges for orders.order_date and customers.signup_date before loading into PostgreSQL, and check that no order precedes its customer's sign-up date.
9. Compare payments.amount with line-item totals per order; if they differ, document which measure is used for revenue.
10. Confirm that no sensitive personal fields (names, e-mail addresses, phone numbers) are present and decide whether any are necessary for the database project.
11. Validate that every proposed analytical query can be answered from the available fields, then freeze the tentative query list and use the confirmed schema for the ER/relational-schema milestone.

6. Source Notes

Dataset source: Kaggle - “Retail Data Warehouse - 12 Table 1M+ Rows Dataset” by Datarspectrum Technology Training Center. The listing describes twelve CSV files.

https://www.kaggle.com/datasets/datarspectrum/retail-data-warehouse-12-table-1m-rows-dataset

Supporting references: the table list, column names, foreign-key structure and approximate row counts were cross-checked against publicly available projects that use this dataset (a PostgreSQL retail-analytics project and a data-profiling project on GitHub). Because these are secondary sources, the team will rely on the actual downloaded files when fixing the final table list, data types and keys.

Course brief: DS604 Introduction to Data Management, Autumn 2026. Milestone 1 requires dataset identification and a tentative list of analytical queries; later milestones require the ER diagram, relational schema and DDL, PostgreSQL loading via Pandas and SQLAlchemy, analytical queries, and views.